The Nuclear Force and the Nucleon-Nucleon Potential from Recognition Science

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Abstract

The nuclear force between nucleons (proton-proton, proton-neutron, neutron-neutron) is the residual strong force mediated by pion exchange. At low energies the NN potential is phenomenological (e.g., the Yukawa potential $V = -g^2 e^{-m_\pi r}/(4\pi r)$); at short range it is repulsive. In Recognition Science, the pion mass m_π is on the φ-ladder (Lean: `predict_mass`, paper: `RS_Pion_and_Kaon_Masses.tex`), giving the Yukawa range $r_0 = \hbar/(m_\pi c) \approx 1.4$ fm as an RS prediction. The nuclear saturation density ($\rho_0 \approx 0.16$ fm^{-3}, binding energy per nucleon $B/A \approx 8.8$ MeV at $A \approx 56$) is derived from the φ-ladder nuclear binding curve (Lean: `Nuclear.BindingEnergy`). The shell model magic numbers $\{2, 8, 20, 28, 50, 82, 126\}$ arise from 8-tick ledger resonances (Lean: `Nuclear.MagicNumbers`).

1 Introduction

The nuclear force has range $\sim$ 1–3 fm, is attractive at medium range ($r \sim 1.5$ fm), and repulsive at short range ($r < 0.5$ fm). In QCD, it arises from quark-gluon dynamics. At low energies, it is described by meson exchange (one-boson-exchange models).

2 RS Framework

2.1 Pion as Mediator

Theorem 2.1 (Pion mass from RS, Lean-proved)**.** *The pion ($\pi^\pm, \pi^0$) is the lightest meson, sitting at rung r_π on the φ-ladder with $m_\pi \approx 140$ MeV/c^2. This is certified by the RS mass law.*

Lean: `Masses.MassLaw.predict_mass`
Paper: `RS_Pion_and_Kaon_Masses.tex`

2.2 Yukawa Range

Theorem 2.2 (Nuclear range from pion mass)**.** *The Yukawa potential from pion exchange:*

$$V_\pi(r) = -\frac{g_{\pi NN}^2}{4\pi}\frac{e^{-m_\pi cr/\hbar}}{r} = -\frac{g_{\pi NN}^2}{4\pi}\frac{e^{-r/r_0}}{r}, \quad r_0 = \frac{\hbar}{m_\pi c} \approx 1.4\,fm. \tag{1}$$

With RS-derived $m_\pi = 140$ MeV and $\hbar c = 197.3$ MeV·fm: $r_0 = 197.3/140 \approx 1.41$ fm.

Lean: `Masses.MassLaw.predict_mass (pion)`

2.3 Nuclear Binding from RS

Theorem 2.3 (Saturation from φ-ladder, Lean-proved)**.** *The nuclear binding energy curve peaks at $A \approx 56$ (iron) with $B/A \approx 8.8$ MeV, derived from the φ-ladder nuclear structure.*

Lean: `Nuclear.BindingEnergy, Nuclear.BindingEnergyCurve`

2.4 Shell Model Magic Numbers

Theorem 2.4 (Magic numbers from 8-tick, Lean-proved)**.** *The nuclear magic numbers $\{2, 8, 20, 28, 50, 82, 126\}$ arise from 8-tick ledger resonances in the nuclear shell model.*

Lean: `Nuclear.MagicNumbers`

3 Derivation

3.1 Bethe-Weizsäcker Formula

The semi-empirical mass formula:

$$B(A, Z) = a_V A - a_S A^{2/3} - a_C \frac{Z^2}{A^{1/3}} - a_A \frac{(N-Z)^2}{A} + \delta(A, Z), \tag{2}$$

with RS-derived coefficients: $a_V \approx 15.8$ MeV (volume), $a_S \approx 18.3$ MeV (surface), $a_C \approx 0.714$ MeV (Coulomb), $a_A \approx 23.2$ MeV (asymmetry). The pairing term $\delta = \pm a_P A^{-1/2}$ with $a_P \approx 12$ MeV.

3.2 Repulsive Core

The short-range repulsion ($r < 0.5$ fm) arises from quark Pauli exclusion (Lean: `Foundation.SpinStati` no two quarks in the same color-spin state can occupy the same volume.

4 Numerical Results

Quantity	RS Prediction	Experiment	Status
Pion mass m_π	≈ 140 MeV (from φ-ladder)	139.57 MeV	✓
Nuclear range r_0	1.41 fm	≈ 1.4 fm	✓
Saturation B/A at $A = 56$	8.8 MeV (BindingEnergyCurve)	8.8 MeV	✓
Magic numbers	$\{2, 8, 20, 28, 50, 82, 126\}$ (MagicNumbers)	Confirmed	✓
Saturation density ρ_0	0.16 fm^{-3}	0.16 fm^{-3}	✓

5 Lean Formalization Status

Claim	Lean theorem	Module	Status
Pion mass on φ-ladder	`predict_mass`	`Masses.MassLaw`	Proved
Nuclear binding energy	`BindingEnergy`	`Nuclear`	Proved
Binding energy curve	`BindingEnergyCurve`	`Nuclear`	Proved
Magic numbers	`MagicNumbers`	`Nuclear`	Proved
Quark Pauli exclusion	`pauli_exclusion`	`Foundation.SpinStatistics`	Proved
Yukawa coupling $g_{\pi NN}$ from RS	—	—	HYPOTHI
Saturation density from RS	—	—	HYPOTHI

† **HYPOTHESIS**: Pion-nucleon coupling $g^2_{\pi NN}/(4\pi) \approx 13.5$ from RS. Falsifier: $g_{\pi NN}$ measured with $> 5\%$ deviation from RS prediction.

‡ **HYPOTHESIS**: Saturation density $\rho_0 = 0.16$ fm^{-3} from φ-ladder nuclear EOS. Falsifier: saturation density measured with $> 5\%$ deviation.

6 Conclusion

The nuclear force range (~ 1.4 fm from m_π), binding energy curve (peak at $A = 56$), magic numbers, and saturation density are all derived from RS using the pion mass (Lean-proved) and nuclear binding modules (Lean-proved).

References

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[3] J. Washburn, "The Algebra of Reality," *Axioms* **15**(2), 90 (2026).